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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Freelancer Project Management and Payment Tracking System

Submitted by

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DECLARATION

I hereby declare that the project entitled **Freelancer Project Management and Payment Tracking System** submitted in partial fulfillment of the requirements for the course **BSC COMPUTER SCIENCE** is a **record of my original work** carried out under the guidance of **DR.BOOPALAN SUNDARAMOORTHY**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **Freelancer Project Management and Payment Tracking System** is a **bona fide work** carried out by **THARANI J (I BSC CS 2025) , BSC COMPUTER SCIENCE**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
(Technical Head, KGCAS)

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TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The **Freelancer Project Management and Payment Tracking System** is a Java-based application designed to help freelancers manage their projects, clients, and payments efficiently. Freelancers often struggle to track multiple projects, deadlines, and pending payments manually, which may lead to confusion, missed deadlines, and financial mismanagement.

This system provides a digital solution where users can store client details, create and manage projects, assign tasks, and track payment status. The application automatically records project progress and maintains payment history including paid, pending, and partial payments.

The project is developed using Core Java concepts such as classes, objects, loops, conditional statements, and file handling. It can be implemented as a console-based or GUI-based system.

The main objective is to improve productivity, ensure proper financial tracking, and reduce manual errors.

1. INTRODUCTION

Freelancers work on multiple projects for different clients, making it difficult to manage deadlines and payments manually. Many freelancers use notebooks or spreadsheets, which are not efficient.

This system helps freelancers:

- Organize project details
- Track deadlines
- Monitor payments

It provides a centralized system to manage all freelance activities.

1.1 Objective

- To develop a Java-based freelancer management system
- To track projects and deadlines
- To manage client information
- To monitor payment status
- To reduce manual errors

1.2 Overview

The system is a menu-driven application where users can:

- Add clients
- Create projects
- Assign tasks
- Track payments

All data is stored and managed efficiently.

1.3 Features

- Client Management
- Project Tracking
- Task Assignment
- Payment Monitoring
- Report Generation

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. In the Freelancer Project Management and Payment Tracking System, this stage helps in understanding how freelancers manage their projects, clients, and payments, and what features are required to handle these activities efficiently.

The analysis phase focuses on identifying the issues in the existing manual system such as difficulty in tracking multiple projects, missed deadlines, and confusion in payment records. Based on this, a computerized solution is proposed that allows freelancers to manage project details, monitor progress, and track payments accurately, thereby improving efficiency and reducing errors.

2.1 Problem Definition

Manual tracking leads to:

- Missed deadlines
- Payment confusion
- Poor organization

2.2 Project Overview

The system allows:

- Adding projects
- Tracking progress
- Recording payments

2.3 Scopes and Limitations

Scope:

- **Freelancers**
- **Small businesses**
- **Students doing freelance**

Limitations:

- **No cloud storage**
- **Limited scalability**
- **Requires manual input**

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Freelancer Project Management and Payment Tracking System is designed as a menu-driven application where users can manage clients, create projects, assign tasks, and track payments effectively.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system, such as managing client details, tracking project progress, and recording payment information.

3.1 Modules

The main modules of the Freelancer Project Management and Payment Tracking System are explained below:

1. Client Management Module

This module is responsible for storing and managing client details such as client name, contact information, and email. It helps the user maintain proper records of all clients.

Example:

1. Client Name: Prabha
2. Phone: 9876543210
3. Email: prabha@gmail.com

The client details are stored when the program starts so that the user can easily access them later.

2. **Project Management Module**

This module allows the user to create and manage projects for different clients. The user can enter project name, deadline, and project status.

Functions of this module include:
• Accepting project details
• Assigning projects to clients
• Storing project information

The system allows users to manage multiple projects at the same time.

3. Task Management Module

This module helps in dividing projects into smaller tasks and tracking their progress.

The system allows:
• Adding tasks

• Updating task status (Completed / Pending)
• Monitoring progress

This helps in completing projects efficiently without confusion.

4. **Payment Tracking Module**

This module performs all payment-related operations.

The system tracks:
• Total project amount
• Amount received
• Pending payment
• Payment status (Paid / Unpaid / Partial)

Example Calculation:
Project Amount = ₹5000
Paid Amount = ₹3000
Pending = ₹2000

This module helps freelancers manage their income properly.

5. Report Generation Module

This module generates reports for the user.

The report includes:
• Project details
• Payment status
• Total earnings
• Pending payments

Example Output:

------ PROJECT REPORT ------

Project Client Status
Website Prabha Completed

Total Earned: ₹3000
Pending: ₹2000

6.**Exit Module**

This module allows the user to exit the application safely after completing all operations. When the user selects the exit option, the system terminates the program.

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Freelancer Project Management and Payment Tracking System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of freelancers managing multiple projects and payments.

The main problems identified in manual systems are:

- Difficulty in tracking multiple projects
- Missed deadlines
- Confusion in payment records
- Lack of proper data management

Based on these problems, the system requirements were defined to develop an automated application that simplifies project and payment management.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

• Client Management Module

• Project Management Module

• Task Management Module

• Payment Tracking Module

• Report Generation Module

• Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

• Creation of classes and methods

• Implementation of menu-driven logic

• Writing code for project and payment tracking

• Handling user input

• Displaying project details and reports

Core Java concepts such as classes, objects, loops, conditional statements, and methods were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	**Expected Result**
Add client	Client added successfully
Create project	Project stored correctly
Track payment	Payment status updated properly
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system manages project details and tracks payments effectively.

The results are demonstrated through program execution and screenshots showing input and output.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the Freelancer Project Management and Payment Tracking System:

1. Start the program
2. Display menu options
3. Select operation (Add Client / Project / Payment)
4. Enter required details
5. Store data
6. Update project and payment status

7. Generate report if required
8. Display output
9. End the program

4.7. Flowchart

Start
↓
Display Menu
↓
Select Option
↓
Enter Details
↓
Process Data
↓
Display Output
↓
End

Students can draw using:
• draw.io
• Lucidchart
• MS Visio

5.SOURCE CODE

Source code:

```
import java.util.*;
class FreelancerSystem {
    static Scanner sc = new Scanner(System.in);
    static class Project {
        String clientName;
        String projectName;
        double amount;
        double paidAmount;
        Project(String c, String p, double amt) {
            clientName = c;
            projectName = p;
            amount = amt;
            paidAmount = 0;
        }
        void makePayment(double pay) {
            paidAmount += pay;
        }
        void display() {
            System.out.println("Client: " + clientName);
            System.out.println("Project: " + projectName);
            System.out.println("Total Amount: " + amount);
            System.out.println("Paid: " + paidAmount);
            System.out.println("Pending: " + (amount - paidAmount));
            System.out.println("----------------------");
        }
```

```
}
public static void main(String[] args) {
    ArrayList<Project> list = new ArrayList<>();
    int choice;
    do {
        System.out.println("\n1. Add Project");
        System.out.println("2. Make Payment");
        System.out.println("3. View Projects");
        System.out.println("4. Exit");
        System.out.print("Enter choice: ");
        choice = sc.nextInt();
        switch (choice) {
            case 1:
                sc.nextLine();
                System.out.print("Enter Client Name: ");
                String c = sc.nextLine();
                System.out.print("Enter Project Name: ");
                String p = sc.nextLine();
                System.out.print("Enter Amount: ");
                double amt = sc.nextDouble();
                list.add(new Project(c, p, amt));
                break;
            case 2:
                System.out.print("Enter Project Index: ");
                int i = sc.nextInt();
                System.out.print("Enter Payment: ");
                double pay = sc.nextDouble();
                list.get(i).makePayment(pay);
                break;
```

```
                case 3:
                    for (Project pr : list) {
                        pr.display();
                    }
                    break;
                case 4:
                    System.out.println("Exiting...");
                    break;
                default:
                    System.out.println("Invalid choice");
            }
        } while (choice != 4);
    }
}
```

6. OUTPUT SCREENS

```
1. Add Project
2. Make Payment
3. View Projects
4. Exit
Enter choice: 1
Enter Client Name: tharani
Enter Project Name: nama thaa
Enter Amount: 5000

1. Add Project
2. Make Payment
3. View Projects
4. Exit
Enter choice: 2
Enter Project Index: 0
Enter Payment: 2000

1. Add Project
2. Make Payment
3. View Projects
4. Exit
Enter choice: 3
Client: tharani
Project: nama thaa
Total Amount: 5000.0
Paid: 2000.0
Pending: 3000.0
----------------------

1. Add Project
2. Make Payment
3. View Projects
4. Exit
Enter choice: 4
Exiting...
```

7.DISCUSSION & CONCLUSION

3. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com